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Lighting device assembly with driver rail mount system

Abstract

The arrangements disclosed herein relate to a lighting device assembly includes an optic assembly including a light source, a housing including a front side, a back side, and an opening in the front side, the front side, the back side, and the opening defining a cavity in which the at least a portion of the optic assembly is received, and the housing is configured to secure the lighting device assembly to a wall, a ceiling, or a surface, and a rail mount system coupled to the back side of the housing. The rail mount system includes a rail coupled to the back side of the housing, wherein the rail includes a lead-in portion, a middle portion, and an end portion, wherein the lead-in portion is recessed from the middle portion, the end portion includes a first secure member, and a plate configured to support a driver configured to drive the light source, the plate includes a side flange. The middle portion of the rail forms a channel with the back side of the housing. The side flange is configured to be received in and slidably coupled to the channel, wherein the side flange includes a second secure member at an end of the side flange. The first secure member and the second secure member are configured to engage one another to position the plate at a predetermined position on the back side of the housing.

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Background/Summary

BACKGROUND

(1) Certain lighting devices such as, but not limited to, room or area lighting devices, can include configurations that allow for mounting of the lighting device in a recess in a ceiling, wall, or another structure. In certain contexts, it can be desirable to mount the lighting device assembly behind a panel of the ceiling, wall, or another structure, and reduce or minimize the size of an opening through the panel for passing light from the lighting device.

(2) Lighting devices include a light source, such as a Light Emitting Diode (LED). Typically, the brightness of an LED light source is at least partially related to the speed in which heat can be transferred away from the LED component. For example, it can be desirable to maintain the temperature of the LED under about 105° Celsius for improved or maximum light output and efficiency. However, in contexts in which the lighting device is mounted in a ceiling, wall or other object (as in the case of a recessed lighting device), the LED component can be located within an enclosed or poorly ventilated environment within the ceiling, wall or other object, which can inhibit the ability to transfer heat away from the LED. In addition, in contexts in which the lighting device is mounted in a ceiling, wall or other object (as in the case of a recessed lighting device), it can be desirable to provide access to components of the lighting device, during or after mounting the lighting device, e.g., in a plenum, attic space, wall space or other volume space in the ceiling, wall or other object.

SUMMARY

(3) Various lighting device and system examples described herein provide efficient transfer and dissipation of heat away from the LED and other heat generating components such as the driver for the LED, ease of accessibility to components located in a ceiling, wall or other object, ability to mount components and pass light through a relatively small opening in a ceiling, wall or other

object, and ease of assembling, installation, disassembling, and removal of the lighting device and system.

(4) In some arrangements, a lighting device assembly includes an optic assembly including a light source, a housing including a front side, a back side, and an opening in the front side, the front side, the back side, and the opening defining a cavity in which the at least a portion of the optic assembly is received, and the housing is configured to secure the lighting device assembly to a wall, a ceiling, or a surface, and a rail mount system coupled to the back side of the housing. The rail mount system includes a rail coupled to the back side of the housing, wherein the rail includes a lead-in portion, a middle portion, and an end portion, wherein the lead-in portion is recessed from the middle portion, the end portion includes a first secure member, and a plate configured to support a driver configured to drive the light source, the plate includes a side flange. The middle portion of the rail forms a channel with the back side of the housing. The side flange is configured to be received in and slidably coupled to the channel, wherein the side flange includes a second secure member at an end of the side flange. The first secure member and the second secure member are configured to engage one another to position the plate at a predetermined position on the back side of the housing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects and features of the present invention will become more apparent to those skilled in the art from the following detailed description of the example embodiments with reference to the accompanying drawings, in which:

(2) FIG. 1A is a front perspective view of an example of a lighting device assembly, according to various embodiments.

(3) FIG. 1B is a back perspective view of an example of a lighting device assembly, according to various embodiments.

(4) FIG. 1C is a front view of an example of the lighting device assembly, according to various embodiments.

(5) FIG. 1D is a back view of an example of the lighting device assembly, according to various embodiments.

(6) FIGS. 1E and 1F are side views of an example of the lighting device assembly, according to various embodiments.

(7) FIG. 2 is a perspective view of a portion of a housing of the lighting device assembly and an adjustment mechanism for adjusting a position of the optic assembly of the lighting device assembly, according to various embodiments.

(8) FIG. 3A shows a first side of an example adjustment member for adjusting the collar and the optic assembly relative to the housing, according to some embodiments.

(9) FIG. 3B illustrates a second side of an example adjustment member for adjusting the collar and the optic assembly relative to the housing, according to some embodiments.

(10) FIG. 3C illustrates an example adjustment member for adjusting the collar and the optic assembly relative to the housing, including a partial cross-sectional view of a fastener and a first groove, according to some embodiments.

(11) FIG. 4A illustrates a first position of a collar relative to the housing, according to some embodiments.

(12) FIG. 4B illustrates a second position of the collar relative to the housing, according to some embodiments.

(13) FIG. 5A is a front perspective view of an example of the optic assembly, according to various embodiments.

- (14) FIG. 5B is a back perspective view of an example of the optic assembly, according to various embodiments.
- (15) FIGS. 5C and 5D are side views of an example of the optic assembly, according to various embodiments.
- (16) FIG. 5E is a cross-sectional view of an example of the optic assembly, according to various embodiments.
- (17) FIG. 5F is an exploded view of an example of the optic assembly, according to various embodiments.
- (18) FIG. 6A is a front perspective view of an example of a light module of the optic assembly, according to various embodiments.
- (19) FIG. 6B is a back perspective view of an example of the light module, according to various embodiments.
- (20) FIGS. 6C and 6D are side views of an example of the light module **310**, according to various embodiments.
- (21) FIG. 6E is a view of an example of the light module with a front portion of the base portion of a light module separated from a back portion of the base portion of a light module **310**, according to various embodiments.
- (22) FIG. 6F is a view of a back portion of the base portion of a light module, according to various embodiments.
- (23) FIG. 6G is a view of a front portion of the base portion of a light module, according to various embodiments.
- (24) FIG. 7A is a perspective view of the back side of the housing with a rail mount system for mounting a driver, according to various embodiments.
- (25) FIG. 7B is a perspective view of a rail of the rail mounting system, according to various embodiments.
- (26) FIG. 7C is a side view of a rail of the rail mounting system, according to various embodiments.
- (27) FIG. 7D is a perspective view of a side flange and a second secure member of a plate of the rail mounting system, according to various embodiments.
- (28) FIG. 7E is a side view of a plate of the rail mounting system, according to various embodiments.
- (29) FIG. 7F is a perspective view of the rail mounting system with a first secure member secured to a second secure member, according to various embodiments.
- (30) FIG. 7G is a perspective view of the rail mounting system with a first secure member secured to a second secure member, according to various embodiments.

DETAILED DESCRIPTION

(31) Hereinafter, example embodiments will be described in more detail with reference to the accompanying drawings. The present invention, however, can be embodied or arranged in various different forms, and should not be construed as being limited to only the illustrated embodiments herein. Rather, these embodiments are provided as examples so that this disclosure will be thorough and complete, and will fully convey the aspects and features of the present invention to those skilled in the art. Accordingly, processes, elements, and techniques that are not necessary to those having ordinary skill in the art for a complete understanding of the aspects and features of the present invention may not be described. Unless otherwise noted, like reference numerals denote like elements throughout the attached drawings and the written description, and thus, descriptions thereof may not be repeated. Further, features or aspects within each example embodiment should typically be considered as available for other similar features or aspects in other example embodiments.

(32) According to various examples described herein, a lighting device assembly or system is configured as a concealed or a recessed lighting device for mounting in a ceiling, wall, surface, or

another structure, by locating at least a portion of the lighting device assembly within or behind a ceiling panel, wall panel, or another structure. For example, the lighting device assembly can be configured to be installed in an opening to a plenum, duct or attic space of a ceiling, or in an inner wall space in a manner to appear flush or substantially flush with an exposed surface of a ceiling, wall, or another object. In other examples, variations of the lighting device assembly can be configured to be installed in a manner that is not flush with an exposed surface (and, instead, are configured to be recessed in or protruding from the exposed surface of a ceiling, wall, outer housing, or another object), or is configured to be surface-mounted on the exposed surface of the ceiling, wall, outer housing or other object. In yet other examples, variations of the lighting device assembly can be configured to be mounted on a support structure (such as, but not limited to a sconce structure, pedestal, shaft or the like).

(33) The lighting device assembly includes a lighting device module (e.g., an optic assembly) having at least one light source for generating light and at least one optic member that are configured to emit the light in a cone or another pattern. In examples in which an optic member includes one or more lenses, the axis of the light emission may correspond to an optical axis of the one or more lenses. In other examples, the axis of the light emission may correspond to a center of the light cone or pattern emitted by the light source and optic member.

(34) Certain examples are configured to provide sufficient thermal communication and heat dissipation characteristics to help maintain the temperature of the light source at or below a desired threshold temperature for improved operation. In addition to thermal communication, the lighting device assembly and the lighting module can be configured for ease of manufacture, assembly, or servicing. In particular examples, the lighting device assembly and the lighting module can be configured to allow adjustment of a direction of light emission from the lighting module about multiple axes.

(35) In some examples, the lighting device assembly can be configured to emit light through a relatively small opening in a panel and (or base member), where that relatively small opening has a size and shape through which the lighting device module and the driver electronics may fit (for example, by installing or removing those components in or from the rest of the lighting device assembly). Accordingly, a single, relatively small opening can provide a light outlet opening, and also accommodate selective access to the lighting device module and (or) the driver electronics, without requiring removal of the rest of the lighting device assembly from an installed state.

(36) **Lighting Device Assembly 100**

(37) FIG. 1A is a front perspective view of an example of a lighting device assembly **100**, according to various embodiments. FIG. 1B is a back perspective view of an example of the lighting device assembly **100**, according to various embodiments. FIG. 1C is a front view of an example of the lighting device assembly **100**, according to various embodiments. FIG. 1D is a back view of an example of the lighting device assembly **100**, according to various embodiments. FIGS. 1E and 1F are side views of an example of the lighting device assembly **100**, according to various embodiments.

(38) The lighting device assembly **100** in FIGS. 1A-1F is shown in an assembled state. The lighting device assembly **100** can be attached to or installed on a panel (e.g., a ceiling panel, a wall panel, or a panel of another structure) in which the lighting device assembly **100** is installed or configured to be installed. The lighting device assembly **100** can be attached to or installed in a plenum, attic space, wall space, or another volume space in the ceiling, wall, surface, or another object. An individual viewing the lighting device assembly **100** when the lighting device assembly **100** is attached to or installed on the panel can observe the features (e.g., the optic assembly **110**) of the lighting device assembly **100** that are not covered by the panel. The lighting device assembly **100** includes an optic assembly **110**, a housing **120**, and brackets **130a** and **130b**.

(39) As used herein, a front direction is a direction in which an optic assembly **110** (e.g., a light source) of the lighting device assembly **100** faces, a direction of the light emission from the light

source, and so on. A front side or a front surface face in the front direction. A back direction is a direction opposite to the front direction. A back side or a back surface face in the back direction. In some examples, the front direction and the back direction are opposite directions along an optical axis of the one or more lenses of the lighting device assembly **100**. A lateral side or surface extends between the front side or surface and the back side or surface. In some examples, the area for which the optic assembly **110** is arranged to provide lighting is in the front direction relative of the lighting device assembly **100**. In some examples, the back side, back surface, lateral side, or lateral surface of the lighting device assembly **100** are at least partially enclosed by the plenum, attic space, wall space, or another volume space in the ceiling, wall, or another object. As used herein a vertical axis is an axis that is or is parallel to the optical axis of the one or more lenses of the lighting device assembly **100**. A horizontal axis is an axis that is perpendicular or transverse to the vertical axis. Extending along an axis or dimension refers to extending along a given axis or dimension or along another axis or dimension that is parallel to the given axis or dimension.

(40) The housing **120** includes a front side **121**, a back side **122**, and a plurality of lateral sides **123**. In some examples as shown, the housing **120** has a generally flat, rectangular cuboid, box-like shape. In other examples, the housing **120** may have other suitable shapes or configurations. The housing **120** (e.g., the sides **121**, **122**, and **123** thereof) can be made of any suitably rigid material and, in particular examples, is made of a material having good (relatively high or fast rate) thermal conduction characteristics, such as, but not limited to a heat dissipating metal, plastic, ceramic or composite material, for dissipation of heat from the optic assembly **110** (e.g., the light source), a driver, and any other heat-generating component mounted on or enclosed within the housing **120**. As described in further detail herein, the housing **120** can serve as a heat sink for transferring and dissipating heat from the optic assembly **110**, the driver, and any other heat-generating component to a surrounding environment, such as the plenum, attic space, wall space, or another volume space in the ceiling, wall, or another object adjacent to or contacting the housing **120** or the brackets **130a** and **130b**. In that regard, the housing **120** (e.g., the sides **121**, **122**, and **123** thereof) can include one or more plates of material having an appropriate thickness (e.g., greater than 3 mm or 5 mm) for functioning as a heat sink. In some examples, the housing **120** can be made of an electrically conductive metal material (or other electrically conductive material) that can be electrically connected to ground (e.g., to a ground conductor present at the installation site), to provide a grounded barrier around the components of the lighting device assembly **100**.

(41) The brackets **130a** and **130b** (e.g., hanger bars) are coupled, attached, or fixed to the lateral sides **123** of the housing **120**. For example, the brackets **130a** and **130b** are coupled to the lateral sides **123** via one or more fasteners (e.g., screws, pins, bolts, etc.). Examples of the brackets **130a** include adjustable hanger bars located on different opposite lateral sides **123**. A bracket **130a** includes male and female slippers. The male and female slippers can be expanded or collapsed to mount a lateral side within various sizes of spaces. Examples of the brackets **130b** include butterfly brackets located on different, opposite lateral sides **123**. A bracket **130b** include a middle portion configured to be coupled to a lateral side **123** and a wing on each of two sides of the middle portion. The wing includes holes for receiving one or more fasteners for coupling, attaching, or affixing the wing (along with the lateral side **123**) to a beam, plank, frame, ceiling, wall, surface, or another object in the plenum, attic space, wall space, or another volume space.

(42) Similar to the housing **120**, the brackets **130a** and **130b** can be made of any suitably rigid material and, in particular examples, can be made of a material having good (relatively high or fast rate) thermal conduction characteristics, such as, but not limited to a heat dissipating metal, plastic, ceramic or composite material, for dissipation of heat from the optic assembly **110** (e.g., the light source), a driver, and any other heat-generating component mounted on or enclosed within the housing **120**. In some examples, heat from the housing **120** can be transferred, via conduction, to the brackets **130a** and **130b**, and the housing **120** and the brackets **130a** and **130b** can transfer the heat to the surrounding environment by conduction or convection. In some examples, the

surrounding environment refers to plenum, attic space, wall space, or another volume space in the ceiling, wall, or another object in which the lighting device assembly **100** is installed, as well as the area for which the optic assembly **110** is arranged to provide lighting. In some examples, the brackets **130a** and **130b** can be made of an electrically conductive metal material (or other electrically conductive material) that can be electrically connected to ground (e.g., to a ground conductor present at the installation site), to provide a grounded barrier around the components of the lighting device assembly **100**.

(43) Adjustment Mechanism **200**

(44) FIG. 2 is a perspective view of a portion of the housing **120** and an adjustment mechanism **200** for adjusting a position of the optic assembly **110**, according to various embodiments. The housing **120** includes or defines a cavity **124** in which the at least a portion of the optic assembly **110** is received. For example, the front side **121**, the back side **122**, and the lateral sides **123** can be arranged to define and enclose a space or volume of the cavity **124**. In cavity **124** can store or hold the optic assembly **110**, a driver for driving the optic assembly, wires, cables, and other support structure that support such components.

(45) The housing **120** is configured to secure the lighting device assembly **100** to a wall, a ceiling, or a surface in a plenum, attic space, wall space, or another volume space in the ceiling, wall, or another object. Based on different design requirements and physical dimensions of the installation space (e.g., a plenum, attic space, wall space, or another volume space), a position of the lighting device assembly **100** is adjustable such that a portion of the optic assembly **110** can be exposed from the installation space to appear flush or substantially flush with an exposed surface of a ceiling, wall, or another object. The adjustment mechanism **200** can allow a user to manually adjust the length of extension of the optic assembly **110** extending from the front side **121** of the housing **120** along a vertical axis with efficiency and accuracy, and in some cases without any tools, to facilitate quick deployment of the lighting device assembly **100** during the installation process. The vertical axis is perpendicular to the front side **121** and/or the back side **122** and parallel to one or more lateral sides **123**. A horizontal axis is perpendicular or transverse to the vertical axis. For example, to adjust for a thicker wall or wider plenum along the vertical direction, the adjustment mechanism **200** can be used to extend the optic assembly **110** farther relative to the front side **121**, whereas to adjust for a thinner wall or narrower plenum along the vertical direction, the adjustment mechanism **200** can be used to retract the optic assembly **110** closer to the front side **121** or more into the cavity **124**.

(46) The adjustment mechanism **210** includes a collar **210** coupled to an opening **125** of the housing **120** (e.g., of the front side **121**) that defines the cavity **124**. For example, the collar **210** includes lateral sides or walls **211a**, **211b**, **211c**, and **211d** that enclose the opening **125** and define a space for receiving the optic assembly **110**. The lateral sides **211a**, **211b**, **211c**, and **211d** have inner surfaces that face the optic assembly **110** when the optic assembly **110** is coupled to the collar **210**. The cavity **124** is accessible through the opening **125**. As shown, the opening **125** and the collar **210** may be offset from a center of the front side **121** of the housing **120**. Although the opening **125** and the collar **210** are shown to have a rectangular shape, other suitable shapes of the opening **125** and the collar **210** that conform to a shape of the cover or housing of the optic assembly **110** can be likewise implemented. The collar **210** can be adjustably coupled, supported, or attached to the front side **121** of the housing **120** at or adjacent to the opening **125**. The optic assembly **110** can be inserted into an interior space of the collar **210** by a user. The optic assembly **110** is removably coupled to the collar **210**. An orientation of the light source in the optic assembly **110** is configured to adjustable by pivoting the light source within the optic assembly **110**.

(47) The collar **210** is fixed relative to the optic assembly **110** when the optic assembly **110** is coupled to the collar **210** and is configured to be adjusted relative to the housing **120** (e.g., to the front side **121**). In other words, the position of the optic assembly **110** relative to the housing **120** can be adjusted by adjusting the position of the collar **210** relative to the housing. By providing an

intermediate component such as the collar **210** to facilitate adjustment of the optic assembly **110**, the lighting device assembly **100** can use any type of optic assembly **110** (e.g., any size or shape of the optic assembly **110**, any number of light sources, etc.) including off-the-shelf optic assemblies and especially-designed optic assemblies for specific purposes and functions.

(48) The collar **210** includes or is coupled to an adjustment member **220** that is configured to adjust, relative to the housing **120** (e.g., the front side **121**), at least one of a position of the collar **210**, a position of the optic assembly **110**, or a position of the light source in the optic assembly **110**. In some examples, a first portion of the collar **210** is within the cavity **124** and a second portion of the collar **210** is external to the cavity **124** and extends beyond the front side **121** (e.g., the front surface of the front side **121**). In some examples, another portion of the optic assembly **110** is external to the cavity **124** and extends beyond the front side **121** (e.g., the front surface of the front side **121**).

(49) FIG. 3A shows a first side of an example adjustment member **220** for adjusting the collar **210** and the optic assembly **110** relative to the housing **120**, according to some embodiments. FIG. 3B illustrates a second side of an example adjustment member **220** for adjusting the collar **210** and the optic assembly **110** relative to the housing **120**, according to some embodiments. FIG. 3C illustrates an example adjustment member **220** for adjusting the collar **210** and the optic assembly **110** relative to the housing **120**, including a partial cross-sectional view of the fastener **232** and a first groove **230**, according to some embodiments.

(50) The first side of the adjustment member **220** and the second side of the adjustment member **220** are opposite sides of the adjustment member **220**, which is located on or adjacent to the lateral side **211a** of the collar **210**. For example, FIG. 3A shows the first side of the adjustment member **220** on an inner surface **212a** of the lateral side **211a** of the collar **210**, and FIG. 3B shows the second side of the adjustment member **220** on an outer surface **213a** of the lateral side **211a** of the collar **210**. The inner surface **212a** of the lateral side **211a** faces the optic assembly **110** when the optic assembly **110** is coupled to the collar **210**. The outer surface **213a** faces a direction opposite to the inner surface **212a** and faces away from the optic assembly **110** when the optic assembly **110** is coupled to the collar **210**. The same adjustment member **220** is located on or adjacent to the lateral side **211b** of the collar **210**.

(51) The adjustment member **220** includes a first groove **230** on the lateral side **212a** that extends along a vertical axis of the lateral side **212a**. The first groove **230** is an elongated, oblong cutout from the lateral side **212a**. In some examples, the first groove **230** is located in the center or substantially the center of the lateral side **212a**. The adjustment member **220** further includes a guide plate **231** and a fastener **232**. The guide plate **231** is slidably coupled to the lateral side **212a**. The fastener **232** is configured to fasten the guide plate **231** to the first groove **230** at a location along the vertical axis or the vertical dimension of the first groove **230**. The guide plate **231** includes a hole, which is configured to be aligned to (e.g., checks in, mates with, combines with, slots into, etc.) the first groove when the guide plate **231** is slidably coupled to the lateral side **212a**. The fastener **232** extends through the hole and the first groove **230** at the location along the vertical axis of the first groove **230** to fasten the guide plate **231** to the first groove **230** at that location.

(52) Examples of the fastener **232** include at least one of a pin, a screw (e.g., a winged screw, a thumb screw), a nut (e.g., a winged nut, a thumb nut), a threaded stud, an insert, or a bolt. In some examples, the fastener **232** allows a user to fasten the guide plate **231** without the use of tools in view of the small and narrow opening **125**. The hand of the user can reach fastener **232** through the opening **125** (before the optic assembly **110** is inserted into the collar **210**) to tighten the fastener **232** at a desired location along the vertical axis of the first groove **230**. In that regard, as shown, the fastener **232** includes a winged screw with the wing portion on the inner surface **212a** of the lateral side **211a** and protruding into the opening **125**. The corresponding nut is located on the outer surface **213a** of the lateral side **211a** to receive the threaded portion of the winged screw. In other examples, the fastener **232** includes a winged nut with the wing portion on the inner surface **212a**

of the lateral side **211a** and protruding into the opening **125**. The corresponding threaded bolt is located on the outer surface **213a** of the lateral side **211a** to be threaded into the winged nut. Other examples of fasteners **232** that allows easy manual fastening or fastening using tools (e.g., a tong or a screwdriver) can be likewise implemented.

(53) The adjustment member **220** includes one or more grooves (e.g., a second groove **242** and a third groove **243**) each extending along a vertical axis of the lateral side **212a** of the collar **210**. Each of the second groove **242** or the third groove **243** is an elongated cutout from the lateral side **212a**. In some examples as shown, the grooves **242** and **243** are located on either side of the first groove **230**. In other words, the first groove **230** is between the second groove **242** and the third groove **243** along a horizontal axis perpendicular or transverse to the vertical axis.

(54) The guide plate **231** includes one or more engagement portions (e.g., a first engagement portion **244** and a second engagement portion **245**), each of which is configured to be inserted into a respective one of the second groove **242** or the third groove **243** and slidably coupled to the lateral side **211a** of the collar **210** via the respective one of the second groove **242** or the third groove **243**. As shown, the guide plate **231** includes the engagement portions **244** and **245** on the edges of the guide plate **231** and a middle portion between the engagement portions **244** and **245**. The fastener **232** and the corresponding hole of the guide plate **231** are located in the middle portion of the guide plate **231**. The middle portion of the guide plate **231** appears to be a flat elongated piece of material, and the engagement portions **244** and **245** are bent at an angle (e.g., 90° or approximately 90°) relative to the middle portion, such that the guide plate **231** appears to have an “U” shape. The guide plate **231**, including the middle portion and the engagement portions **244** and **245** can be formed from a single piece of material.

(55) The engagement portions **244** and **245** of the guide plate **231** are each configured to be located at a location along the vertical axis of a respective one of the second groove **242** or the third groove **243**. The location along the vertical axis of the first groove **230** on which the guide plate **231** is fastened (by the fastener **232**) corresponds to (e.g., is aligned with) the locations along the vertical axis of the second groove **242** or the third groove **243** on which the engagement portions **244** and **245** are located. That is, the engagement portions **244** and **245** are configured to be moved (e.g., by sliding) relative to the grooves **242** and **243** to correspond to the movement of the fastener **232** along the first groove **230**, vice versa.

(56) The engagement portions **244** and **245** are bent around the inner surface **212a** of the lateral side **211a** to protrude from the outer surface **213a** of the lateral side **211a** through the grooves **242** and **243**, and are secured at the outer surface **213a** of the lateral side **211a** via respective buckles **246** and **247**. In some examples, the buckles **246** and **247** can extend from a surface of the front side **121**. In some examples, the buckles **246** and **247** can be a portion of the front side **121** that is bent upward at an angle (e.g., 90° or approximately 90°) relative to the rest of the front side **121**. The buckles **246** and **247** secure the engagement portions **244** and **245** and therefore the guide plate **231** at fixed positions relative to the housing **120** (e.g., the front side **121**).

(57) In some examples, a length of the first groove **230** along the vertical axis defines an adjustment range for the at least one of the position of the collar **210**, the position of the optic assembly **110**, or the position of the light source in the optic assembly **110**. At least one of the position of the optic assembly **110**, or the position of the light source in the optic assembly **110** is defined by extension of at least one of the collar **210** or the light source from the housing **120**. For example, the fastener **232** fastening the guide plate **231** at the location corresponding to a first end (e.g., a back end) of the first groove **230** corresponds to a maximum extension of at least one of the collar **210** or the light source from the housing **120**. For example, the fastener **232** fastening the guide plate **231** at the location corresponding to a second, opposite end (e.g., a front end) of the first groove **230** corresponds to a minimum extension of at least one of the collar **210** or the light source from the housing **120**.

(58) FIG. 4A illustrates a first position of a collar **210** relative to the housing **120** (e.g., the front

side **121**) along a vertical axis **270**, according to some embodiments. FIG. 4B illustrates a second position of the collar **210** relative to the housing **120** (e.g., the front side **121**), according to some embodiments. The fastener **232** of the adjustment member **220** fastening the guide plate **231** at a first location of the first groove **230** corresponds to the first position of the collar **210** (and therefore of the light source) from the housing **120** (e.g., the front side **121**), as shown in FIG. 4A. The fastener **232** of the adjustment member **220** fastening the guide plate **231** at a second location of the first groove **230** corresponds to the second position of the collar **210** (and therefore of the light source) from the housing **120** (e.g., the front side **121**), as shown in FIG. 4B. As shown, at the first position of the collar **210**, the collar **210** extends more from the front side **121** of the housing **120** as compared to the collar **210** at the second position. The first location of the first groove **230** on which the guide plate **231** is fastened is closer to the back end of the first groove **230** as compared to the second location of the first groove **230**. The second location of the first groove **230** on which the guide plate **231** is fastened is closer to the front end of the first groove **230** as compared to the first location of the first groove **230**.

(59) The adjustment member **220** includes one or more interference patterns **251** and **252** on the lateral side **211a** extending along the vertical axis of the lateral side **211a**. In some examples, the interference patterns **251** and **252** define a plurality of steps along the vertical axis. As shown, the adjustment member **220** includes a first interference pattern **251** and a second interference pattern **252** on the lateral side **211a** extending along the vertical axis of the lateral side **211a**. The fastener **232** is configured to fasten the guide plate **231** to the first interference pattern **251** at or over at least one of the plurality of steps of the first interference pattern **251** and to the second interference pattern **252** at or over at least one of the plurality of steps of the second interference pattern **252** when the fastener **232** fastens the guide plate **231** to the first groove **230** at the location along the vertical axis of the first groove **230**. The fastener **232** is configured to fasten the guide plate **231** to the interference patterns **251** and **252** at or over at least one of the plurality of steps when the fastener **232** fastens the guide plate **231** to the first groove **230** at the location along the vertical axis of the first groove **230**. The first groove **230** is between the first interference pattern **251** and the second interference pattern **252** along a horizontal axis perpendicular or transverse to the vertical axis. The first interference pattern **251** and the second interference pattern **252** are between the second groove **242** and the third groove **243** along the horizontal axis.

(60) In some examples, each of the interference patterns **251** and **252** includes a louver having a plurality of slats, each of the plurality of slats corresponds to one of the plural of steps. In some examples, each of the interference patterns **251** and **252** includes a set of alternating depressions and protrusions arranged along the vertical axis, each of the depressions or each of the protrusions corresponds to one of the plural of steps. As shown, the interference patterns **251** and **252** are formed directly on the lateral side **211a** (e.g., etched, pressed, cut, punched, bent, welded, etc.) to reduce cost and manufacturing time and complexity. In other examples, the interference patterns **251** and **252** can be formed separate from the lateral side **211a** and attached to the lateral side **211a**.

(61) The steps (e.g., slats, depressions, protrusions, etc.) of the interference patterns **251** and **252** can have a regular spacing, such that the distance between two adjacent steps remains the same in the interference patterns **251** and **252**. The total length of the interference patterns **251** and **252** along the vertical axis and the spacing of the interference patterns **251** and **252** can be defined according to standardized dimensions for different design requirements and physical dimensions of the installation space. As such, the steps provides indication of predefined incremental distances for adjusting the position of the collar **210**. In addition to providing visual indication, the steps can also provide tactile feedback for a user adjust the position of the guide plate **231**. As the fastener **232** is loosened and the user can adjust the position of the guide plate **231**, the guide plate **231** contacting a surface of each step (e.g., each protrusion) generates a friction force that temporarily causes additional resistance in moving the guide plate **231** while moving the guide plate **231** across the space between two steps (e.g., each depression) lacks such friction force, causing the user to feel

that it is easier to move the guide plate **231**. The steps can also provide additional friction force when the guide plate **231** is fastened to the steps, thus facilitating a tighter and stronger coupling for the collar **210** and the optic assembly **110**.

(62) The adjustment member **220** includes a back plate **250** slidably coupled to (the outer surface **213a**) the lateral side **211a** of the collar **210**. The guide plate **231** and the back plate **250** are located on opposite surfaces **212a** and **213a** of the lateral side **211a**. The guide plate **231** is coupled to the back plate **250**. At least one of the guide plate **231** or the back plate **250** is fixed relative to the housing **120**. The buckles **246** and **247** which secure the engagement portions **244** and **245** also secure the ends of the back plate **250**, thus affixing the back plate **250** to the guide plate **231**, vice versa. As shown, the buckle **246** secures both the engagement portion **244** and one end of the back plate **250**, and the buckle **247** secures both the engagement portion **245** and the other end of the back plate **250**. The coupling of the back plate **250** and the guide plate **231** via the buckles **246** and **247** also prevents the collar **210** from being pulled out from the housing **120** through the opening **125**.

(63) In some examples, the back plate **250** is or includes a lock spring, which is a flexible piece of material (e.g., metal) that has a curved shape (a middle portion is curved toward the x direction) and suitable resilience to return to its curved shape. The fastening of the fastener **232** is configured to pull the lock spring toward the interference patterns **251** and **252** of the lateral side **211a** (in a direction opposite to direction x) and pull the lock spring to engage with or contact the patterns **251** and **252** (e.g., the steps thereof). Thus, fastening of the fastener **232** causes the lock spring to flatten. Loosening of the fastener **232** causes the lock spring to return to its curved shape due to its spring force and moving away from the interference patterns **251** and **252** of the lateral side **211a**. In this manner, the lock spring allows the user to easily loosen the fastener **232** without catching the fastener **232**, the guide plate **231**, or the back plate **250** on the steps of the interference patterns **251** and **252** if the fastener **232** is not sufficiently loosened.

(64) In some examples, the lateral side **211a** includes at least one protruding member **261** and **262** (e.g., a clip, indexing feature, slot, guide, etc.) configured to hold the optic assembly **110** when the optic assembly **110** is coupled to the collar **210**. Each of the protruding members **261** and **262** extends from the lateral side **211a** toward the opening **125**. When the optic assembly **110** is inserted into the opening **125**, the external surfaces of the protruding member **260** facing the opening **125** engages or contacts the optic assembly **110** and holds the optic assembly **110** in place via a friction fit. For example, the protruding members **261** and **262**, which has a cantilevered clip shape, bends when the optic assembly **110** comes in contact with the bulge portion of the protruding members **261** and **262**, and the resulting spring force exerts friction force onto the optic assembly **110**. The grooves **230**, **242**, **243**, the interference patterns **251** and **252**, the guide plate **231**, and the fastener **232** are between the protruding members **261** and **262**.

(65) A height in which each of the protruding members **261** and **262** extends toward the opening **125** (e.g., in a direction that is normal to the inner surface **212a** of the lateral side **211a**), even when the protruding members **261** and **262** are deformed when supporting the optic assembly **110**, is greater than a height in which the fastener **232** extends (e.g., in the direction that is normal to the inner surface **212**) toward the opening **125**. The protruding members **261** and **262** having a greater profile into the opening **125** than that of the fastener **232** prevents the collision between the optic assembly **110** and the fastener **232** when the optic assembly **110** is being supported by the protruding members **261** and **262**. In other words, a space is formed between the optic assembly **110** and components including the fastener **232**, the inner surface **212a**, and the guide plate **231** when the optic assembly **110** is being supported by the protruding members **261** and **262**. The opposite lateral side **211b** similarly includes the protruding members **261** and **262**, such that the friction forces from both opposing lateral sides **211a** and **211b** clutch, pitch, or grip the optic assembly **110** in place.

(66) Similar to the housing **120** and the brackets **130a** and **130b**, the collar **210** and the components

thereof can be made of any suitably rigid material and, in particular examples, can be made of a material having good (relatively high or fast rate) thermal conduction characteristics, such as, but not limited to a heat dissipating metal, plastic, ceramic or composite material, for dissipation of heat from the optic assembly **110** (e.g., the light source), a driver, and any other heat-generating component mounted on or enclosed within the housing **120**. In some examples, heat from optic assembly **110** can be transferred, via conduction, to the collar **210**, which in turn transfers the heat to the housing **120**, which in turn transfers the heat to the brackets **130a** and **130b**, and the housing **120** and the brackets **130a** and **130b** can transfer the heat to the surrounding environment by conduction or convection.

(67) Optic Assembly **110**

(68) FIG. 5A is a front perspective view of an example of the optic assembly **110**, according to various embodiments. FIG. 5B is a back perspective view of an example of the optic assembly **110**, according to various embodiments. FIGS. 5C and 5D are side views of an example of the optic assembly **110**, according to various embodiments. FIG. 5E is a cross-sectional view of an example of the optic assembly **110**, according to various embodiments. FIG. 5F an exploded view of an example of the optic assembly **110**, according to various embodiments.

(69) FIG. 6A is a front perspective view of an example of a light module **310** of the optic assembly **110**, according to various embodiments. FIG. 6B is a back perspective view of an example of the light module **310**, according to various embodiments. FIGS. 6C and 6D are side views of an example of the light module **310**, according to various embodiments. FIG. 6E is a view of an example of the light module **310** with a front portion **361** of the base portion **311** of a light module **310** separated from a back portion **362** of the base portion **311** of a light module **310**, according to various embodiments. FIG. 6F is a view of a back portion **362** of the base portion **311** of a light module **310**, according to various embodiments. FIG. 6G is a view of a front portion **361** of the base portion **311** of a light module **310**, according to various embodiments.

(70) The optic assembly **110** includes one or more light modules **310** and a light housing **320**. In some examples, the optic assembly **110** provides a good (relatively high or fast rate) of thermal communication for thermal transfer and dissipation of heat from a light source of each light module **310** to the rest of the light module **310** and the light housing **320**, where the light module **310** and the light housing **320** function as a heat sink when the optic assembly **110** is installed in the lighting device assembly **100**. In some examples, the optic assembly **110** is configured to be selectively installed in and received by (or removed from) the rest of the lighting device assembly **100** (e.g., at the collar **210**), while the rest of lighting device assembly **100** is in a mounted state in or on the ceiling, wall, or another structure via the brackets **130a** and **130b**.

(71) The light housing **320** (e.g., a module housing) defines a receptacle **321** (e.g., an interior volume) that contains and supports other components of the optic assembly **110** such as at least a portion of the light modules **310**. While the light housing **320** is shown to contain and support two light modules **310**, the light housing **320** can be shaped and sized to house and support any number (e.g., one or three or more) of light modules **310**.

(72) The light housing **320** includes lateral sides **322** and a back side **323**. The lateral sides **322** and the back side **323** define and face the receptacle **321**. The back side **323** includes one or more holes **324**, each of which is configured to receive a respective one of the one or more light modules **310**. The light module **310** includes a base portion **311** and a holder portion **312**. The holder portion **312** is a module with a light source **330** (e.g., the holder portion **312** contains and supports a light source **330**).

(73) The holder portion **312** extends through a hole **324** of the back side **323** into the receptacle **321**. The base portion **311** is coupled to the back side **323** of the light housing **320** via one or more first fasteners **313**. Examples of the first fasteners **313** include screws, pins, bolts, etc. In other examples, the first fasteners **313** can include snap clip, latch, clamp, buckle, hook, adhesives, welding, friction fitting, clamps, or other fasteners.

(74) In some examples, one or more of the base portion **311**, the holder portion **312**, the light housing **320** include or function as a heat sink. Each of the base portion **311**, the holder portion **312**, the light housing **320** can be made of any suitably rigid material and, in particular examples, is made of a material having good (relatively high or fast rate) thermal conduction characteristics, such as, but not limited to a heat dissipating metal, plastic, ceramic or composite material, for dissipation of heat from the light source **330**, a driver, and any other heat-generating component mounted on or enclosed within the housing **120**. For example, heat from the light source **330** can be transferred, via conduction, to the holder portion **312** and via conduction or convection, to the housing **120**. The holder portion **312** can transfer the heat, via conduction, to the base portion **311**, which can transfer the heat to the housing via conduction. The base portion **311** and the light housing **320** can transfer the heat, via conduction or convection, to the housing **120**, which transfers the heat to a surrounding environment (directly or via the brackets **130a** and **130b**), such as the plenum, attic space, wall space, or another volume space in the ceiling, wall, or another object adjacent to or contacting the housing **120** or the brackets **130a** and **130b**. In addition, the light housing **320** can transfer heat toward the area to which the light source **330** by convection. That is, the light housing **320** can transfer heat toward air that is in front of the lighting device assembly **100** when the lighting device assembly **100** is installed to a wall, a ceiling, or a surface in a plenum, attic space, wall space, or another volume space in the ceiling, wall, or another object.

(75) In some examples, at least a portion of a front surface **315** of the base portion **311** directly contacts the back side **323** of the light housing **320**. By having the portion of the front surface **315** of the base portion **311** directly contacting the back side **323** of the light housing **320**, heat can transfer from base portion **311** to the light housing **320** by conduction, thus improving the efficiency by which heat is transferred from base portion **311** (and from the light source **330**) to the light housing **320**. In some examples, given that the rate of heat transfer depends on the area of the direct contact, an entirety of the front surface **315** of the base portion **311** directly contacts the back side **323** of the light housing **320**.

(76) In some examples, the back side **323** of the light housing **320** include a groove **325**. The front surface **315** of the base portion **311** includes a ridge **316** having a convex shape that conforms to the concave shape of the groove **325**. The ridge **316** is configured to be received in (e.g., checked in, indexed, or aligned to) the groove **325** when the front surface **315** of the base portion **311** is coupled to the back side **323** of the light housing **320**. The ridge-and-groove configuration allows easy placement and installation of the light module **310** by a user. For example, after the user inserts the holder portion **312** of a light module **310** through a hole **324**, the user would lose view of the portion of the back side **323** of the light housing **320**, thus cannot orient the light module **310** appropriately to align with any threaded holes of the light housing **320** configured to receive the first fasteners **313** for securing the base portion **311** to the back side. The implementation of the groove **325** and the ridge **316** allows the user to orient (e.g., rotate) the base portion **311** after the holder portion **312** is inserted into the hole **324** until the ridge **316** is received and fits into the groove **325**. In the position in which the ridge **316** is received and fits into the groove **325**, the holes in the base portion **311** that receive the first fasteners **313** align with the holes **325**, to allow the first fasteners **313** to fasten to the holes **325**. In some examples, the groove **325** and the ridge **316** extend along a horizontal axis.

(77) In some examples, the base portion **311** is a heat spreader or a heat sink, and has a generally square cuboid shape, a slab shape, a flat shape, and so on. The holder portion **312** has a generally cylindrical shape and protrudes from the front surface **315** of the base portion **311**. In some examples, the slab shape of the base portion **311** (e.g., the front surface **315**) and the height of the cylindrical shape (along the vertical axis) of the holder portion **312** form a right angle. The shape and size of the base portion **311** allow a user to grasp the light module **310** by the base portion **311** with one hand. For example, the size of the base portion is approximately the size of a normal person's hand. Such shape and size of the light module **310** allow easy and fast installation of the

light module **310** to the light housing **320**. Other suitable shapes of base portion **311** and the holder portion **312** can be likewise implemented.

(78) In some examples, the light source **330** is located at or adjacent to a first end of the cylindrical shape of the holder portion **312**. A second end of the cylindrical shape of the holder portion **312** is connected to the base portion **311**. The light source **330** can include any suitable light emitting device or devices. In some examples, each light source **330** includes one or more LEDs or other light source that generates heat during operation. While particular examples described herein include at least one light source **330** having one or more LEDs, other examples of the light source can include other suitable light sources such as halogen, halide, fluorescent, or incandescent light sources, other electrical discharge or electroluminescence device, and so on. In some examples, the light source **330** can be mounted on a circuit board or other support structure, which can be enclosed by the base portion **311**, the holder portion **312**, or by a driver. In some examples, the light source **330** is fixed to and mounted in thermal communication with (e.g., directly contacting through conduction or through convection over a gap) the mounting surface of a heat sink (e.g., the holder portion **312**, as shown in for example FIG. 5E) for the heat sink to efficiently receive and conduct heat from the light source **330**. By dissipating heat away from the light source **330**, the efficiency and light output of the light source **330** can significantly improve. In that regard, in some examples, the holder portion **312** can include a solid or a mostly solid (instead of a hollow) interior.

(79) One or more wires **340** and **341** connect the light source **330** to a connector **342** of the light module **310**. The connector **342** connects the wires **340** and **341** on one end and connects cables to a driver and/or a power source on the other end. In other words, the connector **342**, the wires **340** and **341**, and the cable connects the light source **330** to the driver and/or power source. At least a portion of each of the wire wires **340** and **341** is external to the holder portion **312**. For example, the holder portion **312** can include a groove extending along a vertical axis along a curved external surface of the holder portion **312**. At least a portion of each of the wire wires **340** and **341** is received by and fitted in the groove. In some examples, a holder portion **312** can include two such grooves for receiving two wires **340** and **341**. By running the wires **340** and **341** external to the holder portion **312** instead of internal to the holder portion **312**, the holder portion **312** can remain entirely solid or mostly solid, which improves heat transfer from the light source **330** to the base portion **311** by conduction. The wires **340** and **341** can extend through an internal space of the base portion **311**.

(80) For example, the base portion **311** includes a front portion **361** and a back portion **362**. The front portion **361** includes the front surface **315** of the base portion **311** and a back surface **363** facing the back portion **362**. The front portion **361** includes a first portion **366** and a second portion **367**. The first portion **366** includes the front surface **315** of the base portion **311**. The second portion **367** includes the back surface **363** of the front portion **361** facing the back portion **362**. The back portion **362** includes a front surface **364** facing the front portion **361** (e.g., facing the back surface **363** of the front portion **361**) and a back surface **365** facing away from the front portion **361**. At least a part of the second portion **367** is between the first portion **366** and the back portion **362**. The front portion **361** and the back portion **362** are coupled via one or more second fastener **314**. Examples of the second fasteners **314** include screws, pins, bolts, etc. In other examples, the second fasteners **314** can include snap clip, latch, clamp, buckle, hook, adhesives, welding, friction fitting, clamps, or other fasteners.

(81) The back portion **362** forms a receptacle **354** to receive at least a portion of the front portion **361**, including the second portion **367** and the back surface **363** of the front portion **361**. As shown, the back portion **362** is configured as a cover having a suitable size and shape to encapsulate at least partially the back surface **363** and the side surfaces of the second portion **367**. The back portion **362** functions as a cover to provide access to at least a portion of the wires **340** and **341** and the connector **342**. A user can fasten the fasteners second fasteners **314** to couple the back portion **362** to the front portion **361** to cover and secure the wires **340** and **341** and the connector **342** in

place between the front portion **361** and the back portion **361**, e.g., between the back surface **363** of the front portion **361** and the front surface **364** of the back portion **362**.

(82) The base portion **311** includes at least one chamfered corner. In some examples as shown, the base portion **311** includes 4 chamfered corners, each of which is at a corner of the square or rectangular shape of the base portion **311**. In some examples, each of the front portion **361**, the back portion **362**, the back surface **363**, the front surface **364**, the back surface **365**, the first portion **366**, and the second portion **367** includes a plurality of chamfered corners. The chamfered corners of the base portion **311** allows easy handling (e.g., grasping and placing) of the base portion **311** during the installation and removal processes of the light module **310** with respect to the light housing **320**, provides additional space for accommodating the wires, cables, and connectors **342** while minimizing the spacing between the two adjacent light modules **310**, without impacting the thermal transfer efficiency of the light modules **310**.

(83) The back portion **362** and the second portion **367** are coupled to at least one of the connector **342** or the wires **340** and **341** connected to the connector **342**. The wires **340** and **341** are connected to the light source **330**, and the wires **340** and **341** extends through at least a portion of an internal hole within the volume of at least one of the first portion **366** or the second portion **367**. The second portion **367** has a chamfered corner **380**. The back portion **362** includes a cavity **368** at the chamfered corner **380** of the second portion **367** to receive the connector **342**. For example, the connector **342** can be inserted into and snapped or friction fitted at the cavity **368**. The second portion **367** includes an external space defined by the chamfered corner **380** and a part of the first portion **366**. For example, the external space refers to the space cutout from a would-be portion of the second portion **367** if the second portion **367** has a similar shape and size as that of the first portion **366**, to form the chamfered corner **380**. The chamfered corner **380** has a size greater than the other chamfered corners of the base portion **311**, such as those for the first portion **366**. The at least one of the connector **342** or the wires **340** and **341** is arranged in the external space of the chamfered corner **380** when at least one of the connector **342** or the wires **340** and **341** is coupled to the back portion **362** through the cavity **368**. This allows the at least one of the connector **342** or the wires **340** and **341** to not protrude beyond the profiles (e.g., the back surface **365**), the lateral sides of the base portion **311** (e.g., the lateral sides of the portions **361** and **362**), or the chamfered corners of the base portion **311** (e.g., the chamfered corners of the portions **361** and **362**). Such configuration allows the light modules **310** to have an overall simplistic shape for handling without impacting the thermal transfer efficiency of the light modules **310**. For example, by providing the light modules **310** and the light housing **320** in the manner of described, a user can easily and efficiently install the light module **310** to the light housing **320** and connect the wires and cables using the connectors **342**.

(84) In some examples, the front portion **361** is composed of a single, unitary body of such material, for improved heat dissipating capabilities. In some examples, the body of the front portion **361** is made of a generally solid, unitary piece of material that is configured as described herein. In some examples, the front portion **361** and the holder portion **312** are composed of a single, unitary body of such material, for improved heat dissipating capabilities. In some examples, the body of the front portion **361** and the holder portion **312** is made of a generally solid, unitary piece of material that is configured as described herein. In some examples, the body of the light module **310** can be made by any suitable manufacturing process or processes including, but not limited to molding, machining, extrusion, or combinations thereof. The light module **310** can be made of any suitably rigid material or materials including, but not limited to metal, plastic, ceramic, composite material, or combinations thereof. In particular examples, the light module **310** is made of a material having a good (relatively high or fast rate) of thermal dissipation capabilities such as, but not limited to a heat dissipating metal, plastic, ceramic or composite material.

(85) The light housing **320** includes at least one attachment member **350** in at least one of the lateral sides **322** configured to couple the light housing **320** to a collar **210** of the lighting device

assembly **110** by friction fitting to a surface (e.g., an interior surface) of the collar **210**. In some examples, an attachment member **350** includes a spring clip, a guide, a spring, or a friction member. For example, an attachment member **350**, which has a cantilevered clip shape with a bulge, bends when the inner surfaces of the sides **211c** and **211d** of the collar **210** comes in contact with the bulge portion of the attachment member **350**, and the resulting spring force exerts friction force onto the inner surfaces of the sides **211c** and **211d** of the collar **210**.

(86) When installing the light device assembly **110** into the collar **210**, a user inserts the light device assembly **110** into the collar **210** and the cavity **124** through the opening **125** such that the attachment members **350** move from front edges of the collar **210** to the back edges of the collar **210** (e.g., from front edges of the sides **211c** and **211d** to the back edges of the sides **211c** and **211d**), along the vertical axis (in the back direction), exerting friction force as described along the way. When the attachment members **350** are pushed past the back edges of the sides **211c** and **211d**, a large portion of the lighting device assembly **110**, including the base portion **311** and at least a portion of the holder portion **312**, is received in the cavity **124**, the bulge portions of the attachment members **350** no longer deform due to the lack of contact with the sides **211c** and **211d** and expand. The attachment members **350** would then abut or sit on the back edges of the sides **211c** and **211d** due to gravity and providing resistance for the lighting device assembly **110** to move in the front direction due to gravity.

(87) A user can pull the lighting device assembly **110** from the cavity **124** in a removal process by pulling lighting device assembly **110** (e.g., the light housing **320**) from the housing **120** in the front direction. In the removal process, the attachment member **350** bends when the inner surfaces of the sides **211c** and **211d** of the collar **210** comes in contact with the bulge portion of the attachment member **350**, and the resulting spring force exerts friction force onto the inner surfaces of the sides **211c** and **211d** of the collar **210**. In other example, the attachment member **350** includes a snap clip, latch, clamp, buckle, hook, and so on.

(88) The light housing **320** includes one or more recesses **355** each configured to engage a respective one of the at least one protruding member **261** and **262** of the collar **210**. As shown, an attachment member **350** is provided on a first side of the lateral sides **322** and a recess **355** is provided on a second side of the lateral sides **322**, where the first side is different from the second side. In some examples, at least one recess **355** is provided on one second side of the lateral sides **322**, and at least one recess **355** is provided on another second side of the lateral sides **322**, where the two second sides are opposite sides of the light housing **320**. In some examples, the first side of the lateral sides **322** which has at least one attachment member **350** does not have any recess **355**, and the second side of the lateral sides **322** which has at least one recess **355** does not have any attachment member **350**. When installing the light device assembly **110** into the collar **210**, a user inserts the light device assembly **110** into the collar **210** and the cavity **124** through the opening **125** such that the protruding members **261** and **262** move from the back edge of the second side of the lateral sides **322** toward the front edge of the second side of the lateral sides, along the vertical axis (in the front direction), exerting friction force onto the second side of the lateral sides **322** as described along the way. Each recess **355** functions as a stopper for holding and locking a protruding member **261** or **262** in place when the protruding member reaches the front end of the side **322**. Thus, the recesses **355** are located at or adjacent to the front end of at least one lateral side **322**.

(89) In some examples, the light housing **320** includes a trim member **357**. The trim member **357** may be configured to provide one or more functions including, but not limited to aesthetic or ornamental functions, heat dissipation functions, or combinations thereof. The recesses **355** can be arranged in the trim member **357**, which is located at or around the front end of the lateral sides **322**. The trim member **357** may include a portion for connection to the light housing **320** and a second portion that forms a flange or lip around and adjacent the light outlet opening of the light housing **320**. When the light device assembly **110** is installed in the system **100** and mounted in a

ceiling, wall, surface, or another structure, the flange or lip portion of the trim member **357** can be exposed relative to the exposed surface of a ceiling, wall, outer housing, or another object) of the ceiling, wall, or another structure.

(90) In some examples, the flange or lip portion is configured to cover (and hide from view) a space or gap at the outer circumference of the lateral sides **322** of the light housing **320** and the openings in the mounting surface that may otherwise be visible. Additionally or alternatively, the flange or lip portion can be configured with an ornamental or aesthetic design or an appearance that corresponds to and matches the appearance of the mounting surface to be visually obscure. When installed, the trim member **357** (or the flange or lip of the trim member **357**) may fit flush with or abutted against the mounting surface.

(91) In some examples, an orientation of the light source **330** is configured to adjustable by pivoting the light source **330** within the optic assembly **110**. In some examples, the light housing **320** can include a bezel **326** within the receptacle **321**. While the bezel **326** is generally has a cylindrical shape, other embodiments may include bezel **326** having other suitable shapes, including but not limited to curved or partially spherical shapes, conical, cube or cuboid shapes, rectangular shapes, triangular shapes, or the like.

(92) A rotating member **327** can pivot and/or rotate within the bezel **326** to adjust a direction of emitted light from the light source **330**, which remains fixed relative to the bezel **326** when the optic assembly **110** is installed in the light housing **320**. The rotating member **327** includes an optic member **328** can be a lens, filter, or other optical device that passes light, and affects a characteristic of the light being passed. In some examples, the optic member **328** includes a lens configured to focus light toward one or more focus points or centers of focus. In some examples, the optic member **328** may have a configuration for directing light through a relatively small aperture or opening. Some examples of such optic members that can be employed for optic member **328** are described in the Applicant's U.S. Pat. No. 10,900,654 (which is incorporated herein by reference, in its entirety). In other examples, the optic member **328** may include other suitable lens configurations. The optic member **328** is configured to be pivot and/or rotate with the rotating member **327**.

(93) In some examples, an outer surface of the rotating member **327** can slideably engage a cavity (with a corresponding mating curvature and dimension) of the bezel **326** in a ball and socket manner. Accordingly, in various examples, the rotating member **327** including the optic member **328** can pivot in any direction (e.g., on a 360 degree plane) within the bezel **326**, by slideably engaging the cavity of the bezel **326**. Thus, the outer surface of the rotating member **327** has a curvature (e.g., upper hemisphere portion) that is partially held within the cavity of the bezel **326**.

(94) In some embodiments, at least one of the outer surface of the rotating member **327** or an interior surface of the cavity of the bezel **326** may include a friction member or a friction material coating to provide a friction surface to maintain a pivoted position of the rotating member **327** within the bezel **326**. For example, when the optic member **328** is pressed and pivoted to a desired position relative to the bezel **326** and then released, the elastic member presses the rotating member **327** against the interior surface of the cavity of the bezel **326**, or vice versa, so that the engaging surfaces thereof frictionally engages the friction surface, to prevent or substantially prevent the rotating member **327** from shifting (or sliding) to a different position from the desired position due to gravity (i.e., without manual force) or due to the force exerted by the elastic member. In some examples, the frictional force may be overcome by manual force applied to manually adjust or move (pivot and/or rotate) the rotating member **327** (e.g., by the exposed components of the optic member **328**) relative to the bezel **326**. Accordingly, the friction member or the friction material coating can include any suitable material to provide the friction surface, for example, but not limited to, silicone, rubber, and/or the like. In further examples, the friction surface includes contour, roughness or other features that enhance friction.

(95) Driver Rail Mount System **400**

(96) FIG. 7A is a perspective view of the back side 122 of the housing 120 with a rail mount system 400 for mounting a driver 430, according to various embodiments. FIG. 7B is a perspective view of a rail 410 of the rail mounting system 400, according to various embodiments. FIG. 7C is a side view of a rail 410 of the rail mounting system 400, according to various embodiments. FIG. 7D is a perspective view of a side flange 422 and a second secure member 424 of a plate 420 of the rail mounting system 400, according to various embodiments. FIG. 7E is a side view of a plate 420 of the rail mounting system 400, according to various embodiments. FIG. 7F is a perspective view of the rail mounting system 400 with a first secure member 414 secured to a second secure member 424, according to various embodiments. FIG. 7G is a perspective view of the rail mounting system 400 with a first secure member 414 secured to a second secure member 424, according to various embodiments.

(97) In some arrangements, a rail mount system 400 is operatively coupled to the back side 122 of the housing 120 for removably mounting a driver 430 configured to drive the light source 330. The driver 430 is connected to a power source (e.g., a power outlet, a battery, power line, and so on) via suitable cables. One or more cables from the driver 430 is connected to the connectors 342 for driving the light source 330. The driver 430 is configured to convert power provided by the power source to a suitable power for driving the light source 330. In some examples, the driver 430 can include a processor to execute instructions stored on memory (e.g., non-transient computer readable media) or transmitted to the processor, to process data and/or to control various functions of the lighting device (such as, but not limited to, temperature, light output, color of light, direction of light, focus of light, and/or the like). In particular examples, the light source 330 includes an LED, and the driver 430 includes one or more LED drivers to drive the LED light source 330.

(98) The rail mount system 400 includes at least a rail 410 and a plate 420. The plate 420 is configured to support the driver 430. The driver 430 can be fastened to a front surface of the plate 420 via one or more screws, nails, pins, bolts, etc. In some examples, the plate 420 includes a plurality of holes to receive such screws, nails, and pins to secure drivers 420 of a variety of sizes and shapes to the plate 420. For example, first holes on the plate 420 can be used for securing (e.g., receiving screws, nails, pins, bolts, etc.) a driver of a size and shape, and second holes on the plate 420 can be used for securing (e.g., receiving screws, nails, pins, bolts, etc.) another driver of a different size and shape. In other examples, the driver 430 can be attached to the plate 420 via one or more of a snap clip, latch, clamp, buckle, hook, adhesives, welding, friction fitting, clamps, or other fasteners. The plate 420 is slidably coupled to a front surface 401 of the back side 122 of the housing 120. When the plate 420 and the driver 420 are coupled to the front surface 401 of the back side 122 of the housing 120, and the back side 122 is coupled to the rest of the housing 120 including the lateral sides 123, the driver 420 is within the cavity 124 and facing the optic assembly 110. For example, the driver 420 can be adjacent to the connectors 342 and the light modules 310.

(99) In some examples, the rail 410 is coupled to the backside 122 of the housing 120, e.g., to a front surface 401 of the back side 122 of the housing. The front surface 401 of the back side 122 of the housing faces the optic assembly 110 and the light modules 310. As shown, the rail mount system 400 includes two rails 410 (e.g., a first rail and a second rail), one for each sides of the plate 420. The plate 420 includes side flanges 422 (e.g., a first side flange and a second side flange) located on each of opposing sides of the plate 420. The plate 420 is slidably coupled to the rails 420 via the side flanges 422. That is, the side flanges 422 can slide in a direction y along the rails 420 to couple the plate 420 to the back side 122.

(100) In some examples, each rail 410 includes a lead-in portion 411, a middle portion 412, and an end portion 413. The end portion 413 includes a first secure member 414. A side flange 422 includes a second secure member 424 at an end of the side flange 422. The first secure member 414 is configured to engage with, fasten, mate, or otherwise couple the second secure member 424 when the second secure member 424 slides to the first secure member 424 from the lead-in portion 411, to the middle portion 412, and then to the end portion 413. The lead-in portion 411, middle

portion **412**, and end portion **413** of a rail **410** are joined by a back portion of the rail **410**. The back portion of the rail **410** can affix or attach the rail **410** to the back side **122** via screws, pins, bolts, snap clip, latch, clamp, buckle, hook, adhesives, welding, friction fitting, clamps, or other fasteners. (101) The lead-in portion **411** is configured to initial receive a side flange **422** when a user begins to slide the plate **420** (including the second secure member **424**) in the y direction toward the end portion **413**. The rails **410** extend along a first direction (e.g., the y direction), and the plate **420** is configured to slide against the rail **410** in the first direction. In some examples, the rails **410** on either sides of the backside **122** are parallel to each other, and the side flanges **422** on either side of the plate **420** are parallel to each other.

(102) The distances between two lead-in portions **411** of the two rails **420** may be wider or more open than the middle portions **412** of those rails **420**, to allow the user easy access and alignment of the plate **420** to the rails **420** and the space therebetween. The middle portion **412** of each rail **410** forms a channel **415** with the back side **122** (e.g., the front surface **401** thereof) of the housing **120**. The side flange **422** of the plate **420** is configured to be received in and slidably coupled to the channel **415**. For example, a middle portion **412** of a rail **410** includes a ridge extending from a back portion of the rail **410**. The ridge is spaced apart from the front surface **401** of the back side **122** to form the channel **415**. Thus, the channel **415** is formed between the ridge and the front surface **401** of the back side **122**. A side flange **422** including the second secure member **424** can slide in the channel **415** between the ridge and the back side **401**, after the side flange **422** slides past the lead-in portion **411**. In other words, the lead-in portions **411** of the rails **410** guide the side flanges **422** to the middle portion **412**. For example, at least a portion of the lead-in portion **411** is recessed from the middle portion **412** (e.g., from the ridge).

(103) In some examples, the height of the channel **415** (e.g., the distance between the front surface **401** of the back side **122** and the surface of the ridge facing the front surface **401**) is the same as the thickness of an outer portion of the flange **422** and the second secure member **424**. The ridge of the middle portion **412** (and the rest of the rail **410**) can be made from a suitably rigid and yet slightly flexible material such as a certain plastic, resin, or metal such that when the flange **422** and the second secure member **424** are received in the channel **415**, either sliding across or held in place, the ridge can exert a spring force to push the flange **422** and the second secure member **424** toward the front surface **401** of the back side **122**, such that at least a portion of or the entirety of a surface of the plate **420** abuts or directly contacts the front surface **401** of the back side **122**, for thermal transfer via conduction.

(104) For example, a first side flange **422** is configured to slide against the first rail **410**, from the lead-in portion **411** of the first rail **410** to the middle portion **412** of the first rail **410**, and then to the end portion **413** of the first rail **410**, to engage the second secure member **424** of the first side flange **422** to the first secure member **414** of the first rail **410**. A second side flange **422** is configured to slide against the second rail **410**, from the lead-in portion **411** of the second rail **410** to the middle portion **412** of the second rail **410**, and then to the end portion **413** of the second rail **410**, to engage the second secure member **424** of the second side flange **422** to the first secure member **414** of the second rail **410**.

(105) The end portion **413** of a rail **410** includes the first secure member **414**. The first secure member **414** includes one or more of a snap clip, latch, clamp, buckle, hook, and so on. For example, as shown, the first secure member **414** can include a cantilevered snap clip. The first secure member **414** can be made from a flexible or resilient material such as plastic, resin, Acrylonitrile Butadiene Styrene (ABS), polycarbonate, unfilled nylon, polypropylene, and so on. The second secure member **424** include one or more of a depression, a hole, an indentation, and so on. The first secure member **414** and the second secure member **424** are configured to engage one another to position the plate **420** at a predetermined position on the back side **122** of the housing **120**. The side flange **422** is configured to slide against the rail **410**, from the lead-in portion **411** to the middle portion **412**, and then to the end portion **413**, to engage the second secure member **424**

to the first secure member **414**.

(106) In some examples, the first secure member **414** includes a snap clip, and the second secure member **424** includes a depression configured to receive the snap clip. In some examples, engaging the first secure member **414** and the second secure member **424** includes snap fitting one of the first secure member **414** and the second secure member **424** to another one of the first secure member **414** and the second secure member **424**. The snap clip includes a cantilevered hook, and the depression is configured to receive the cantilevered hook. In some examples, the second secure member **424** includes a protrusion adjacent to the depression. The cantilevered hook catches the protrusion while being received in the depression.

(107) In some examples, when the first secure member **414** and the second secure member **424** engage one another, engaging of the first secure member **414** and the second secure member **424** pushes the plate **420** toward the front surface **401** of the back side **122** of the housing **120** and pushes the front surface **401** of the back side **122** of the housing **120** toward the plate **420**, such that at least a portion of or the entirety of a surface of the plate **420** abuts or directly contacts the front surface **401** of the back side **122**, for thermal transfer via conduction.

(108) The first and second secure members **414** and **424** allow positive engagement of the plate **420** to the back side **122**, and when engaged, can provide resist vibration of the plate **420** relative to the back side **122**, vice versa, and can provide pressure to maintain the direct contact between the plate **420** and the front surface **401** of the back side **122** for thermal conduction. In some examples, the plate **420** can function as or includes a first heat sink, and the back side **122** of the housing **120** includes a second heat sink. The plate **420** directly contacting the back side **122** of the housing **120** allow transfer heat of the driver **430** from the first heat sink to the second heat sink by conduction.

(109) The back side **122**, the plate **420**, and the components thereof can be made of any suitably rigid material and, in particular examples, can be made of a material having good (relatively high or fast rate) thermal conduction characteristics, such as, but not limited to a heat dissipating metal, plastic, ceramic or composite material, for dissipation of heat from the driver **430** and any other heat-generating component mounted on or enclosed within the housing **120**. In some examples, heat from the driver **430** can be transferred, via conduction, to the plate **420**, which in turn transfers the heat to the back side **122** of the housing **120**, which in turn transfers the heat to the brackets **130a** and **130b** and the rest of the housing **120**. The housing **120** and the brackets **130a** and **130b** can transfer the heat to the surrounding environment by conduction or convection.

(110) Given that the lighting device assembly **100** is installed in a plenum, attic space, wall space, or another volume space in the ceiling, wall, or another object, and air flow is very limited or even non-existent in such spaces, primary heat dissipation mechanism for the embodiment described herein relies on heat sinks (e.g., the housing **120**, the brackets **130a** and **130b**, the light housing **320**, the base portion **311**, the holder portion **312**, the plate **410**, and so on) to transfer heat from the light module **330**, the driver **430**, or another heat dissipating component enclosed by the housing **120** to the surround environment.

(111) In the drawings, the relative sizes of elements, layers, and regions can be exaggerated and/or simplified for clarity. Spatially relative terms, such as “beneath,” “below,” “lower,” “under,” “above,” “upper,” and the like, can be used herein for ease of explanation to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or in operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” or “under” other elements or features would then be oriented “above” the other elements or features. Thus, the example terms “below” and “under” can encompass both an orientation of above and below. The device can be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein should be interpreted accordingly.

(112) It will be understood that, although the terms “first,” “second,” “third,” etc., can be used

herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer or section described below could be termed a second element, component, region, layer or section, without departing from the spirit and scope of the present invention.

(113) It will be understood that when an element or layer is referred to as being “on,” “connected to,” “coupled to,” “secured to” or “attached to” another element or feature, it can be directly on, connected to, coupled to, secured to or attached to the other element or layer, or one or more intervening elements or layers can be present. In addition, it will also be understood that when an element or layer is referred to as being “between” two elements or layers, it can be the only element or layer between the two elements or layers, or one or more intervening elements or layers may also be present.

(114) The terminology used herein is for the purpose of describing particular embodiments and is not intended to be limiting of the present invention. As used herein, the singular forms “a” and “an” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and “including,” “has,” “have,” and “having,” when used in this specification, specify the presence of the stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

(115) As used herein, the term “substantially,” “about,” “generally” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent variations in measured or calculated values that would be recognized by those of ordinary skill in the art. Further, the use of “may” when describing embodiments of the present invention refers to “one or more embodiments of the present invention.” As used herein, the terms “use,” “using,” and “used” can be considered synonymous with the terms “utilize,” “utilizing,” and “utilized,” respectively. Also, the term “exemplary” is intended to refer to an example or illustration.

(116) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the present invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and/or the present specification, and should not be interpreted in an idealized or overly formal sense, unless expressly so defined herein.

(117) The foregoing description of illustrative embodiments has been presented for purposes of illustration and of description. It is not intended to be exhaustive or limiting, and modifications and variations can be possible in light of the above teachings or can be acquired from practice of the disclosed embodiments. Various modifications and changes that come within the meaning and range of equivalency of the claims are intended to be within the scope of the invention. Thus, while certain embodiments of the present invention have been illustrated and described, it is understood by those of ordinary skill in the art that certain modifications and changes can be made to the described embodiments without departing from the spirit and scope of the present invention as defined by the following claims, and equivalents thereof.

Claims

1. A lighting device assembly, comprising: an optic assembly comprising a light source; a housing comprising a front side, a back side, and an opening in the front side, wherein the front side, the

back side, and the opening define a cavity in which the at least a portion of the optic assembly is received, wherein the housing is configured to secure the lighting device assembly to a wall, a ceiling, or a surface; a rail mount system coupled to the back side of the housing, the rail mount system comprises: a rail coupled to the back side of the housing, wherein the rail comprises a lead-in portion, a middle portion, and an end portion, wherein the lead-in portion is recessed from the middle portion, the end portion comprises a first secure member; a plate configured to support a driver configured to drive the light source, the plate comprises a side flange, wherein the middle portion of the rail forms a channel with the back side of the housing, wherein the side flange is configured to be received in and slidably coupled to the channel, wherein the side flange comprises a second secure member at an end of the side flange, and wherein the first secure member and the second secure member are configured to engage one another to position the plate at a predetermined position on the back side of the housing.

2. The lighting device assembly of claim 1, wherein the first secure member comprises a snap clip; and the second secure member comprises a depression configured to receive the snap clip.

3. The lighting device assembly of claim 2, wherein the snap clip comprises a cantilevered hook, and the depression is configured to receive the cantilevered hook.

4. The lighting device assembly of claim 3, wherein the second secure member comprises a protrusion adjacent to the depression; and the cantilevered hook catches the protrusion while being received in the depression.

5. The lighting device assembly of claim 1, wherein when the first secure member and the second secure member engage one another: engaging of the first secure member and the second secure member pushes the plate toward the back side of the housing and pushes the backside of the housing toward the plate; and the plate directly contacts the back side of the housing.

6. The lighting device assembly of claim 1, wherein the plate comprises a first heat sink; and the back side of the housing comprises a second heat sink.

7. The lighting device assembly of claim 6, wherein when the first secure member and the second secure member engage one another: the plate directly contacts the back side of the housing; and the first heat sink is configured to transfer heat of the driver from the first heat sink to the second heat sink by conduction.

8. The lighting device assembly of claim 1, wherein the side flange is configured to slide against the rail, from the lead-in portion to the middle portion, and then to the end portion, to engage the second secure member to the first secure member.

9. The lighting device assembly of claim 1, wherein the rail comprises a first rail and a second rail, each of the first rail and the second rail comprises the lead-in portion, the middle portion, the end portion, and the first secure member; and the side flange of the plate comprises a first side flange and a second side flange, each of the first side flange and the second side flange comprises the second secure member.

10. The lighting device assembly of claim 9, wherein the first rail and the second rail are positioned on the back side of the housing to allow the plate to slide between the first rail and the second rail.

11. The lighting device assembly of claim 10, wherein the first side flange is configured to slide against the first rail, from the lead-in portion of the first rail to the middle portion of the first rail, and then to the end portion of the first rail, to engage the second secure member of the first side flange to the first secure member of the first rail; and the second side flange is configured to slide against the second rail, from the lead-in portion of the second rail to the middle portion of the second rail, and then to the end portion of the second rail, to engage the second secure member of the second side flange to the first secure member of the second rail.

12. The lighting device assembly of claim 1, wherein the rail extends along a first direction; and the plate is configured to slide against the rail in the first direction.

13. The lighting device assembly of claim 1, wherein the middle portion of the rail comprises a ridge, the channel being formed between the ridge and the back side of the housing; and the side

flange of the plate is configured to slide in the channel between the ridge and the back side of the housing.

14. The lighting device assembly of claim 1, wherein engaging the first secure member and the second secure member comprises snap fitting one of the first secure member and the second secure member to another one of the first secure member and the second secure member.

15. The lighting device assembly of claim 1, wherein the plate comprises a plurality of holes to receive screws, nails, and pins for a variety of sizes and shapes of the driver.

16. The lighting device assembly of claim 1, wherein comprises wherein the driver comprises at least one of a connector or a wire to connect the light source to a power source.

17. A rail mount system coupled to a back side of a housing, the housing being configured to support an optic assembly comprising a light source, the rail mount system comprises: a rail coupled to the back side of the housing, wherein the rail comprises a lead-in portion, a middle portion, and an end portion, wherein the lead-in portion is recessed from the middle portion, the end portion comprises a first secure member; a plate configured to support a driver configured to drive the light source, the plate comprises a side flange, wherein the middle portion of the rail forms a channel with the back side of the housing, wherein the side flange is configured to be received in and slidably coupled to the channel, wherein the side flange comprises a second secure member at an end of the side flange, and wherein the first secure member and the second secure member are configured to engage one another to position the plate at a predetermined position on the back side of the housing.

18. The rail mount system of claim 17, wherein the plate comprises a heat sink; and when the first secure member and the second secure member engage one another: the plate directly contacts the back side of the housing; and the heat sink is configured to transfer heat of the driver from the heat sink to the back side by conduction.

19. The rail mount system of claim 17, wherein the middle portion of the rail comprises a ridge, the channel being formed between the ridge and the back side of the housing; and the side flange of the plate is configured to slide in the channel between the ridge and the back side of the housing.

20. A rail mount system coupled to a back side of a housing, the housing being configured to support an optic assembly comprising a light source, the rail mount system comprises: a rail coupled to the back side of the housing, wherein the rail comprises a lead-in portion, a middle portion, and an end portion, wherein the lead-in portion is recessed from the middle portion, the end portion comprises a snap clip; a plate configured to support a driver configured to drive the light source, the plate comprises a side flange, wherein the middle portion of the rail forms a channel with the back side of the housing, wherein the side flange is configured to be received in and slidably coupled to the channel, wherein the side flange comprises a depression and a protrusion at an end of the side flange, and wherein a snap fit is formed via the snap clip, the depression, and the protrusion, by the depression receiving the cantilevered hook while the cantilevered hook is catching the protrusion.
